



BACHELOR OF COMPUTER SCIENCE (SOFTWARE ENGINEERING)

COURSE NAME AND CODE:

CSE3023 – WEB-BASED APPLICATION DEVELOPMENT (K1)

LAB 1

SEMESTER II SESSION 2025/2026

PREPARED BY:

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LAB MODERATOR:

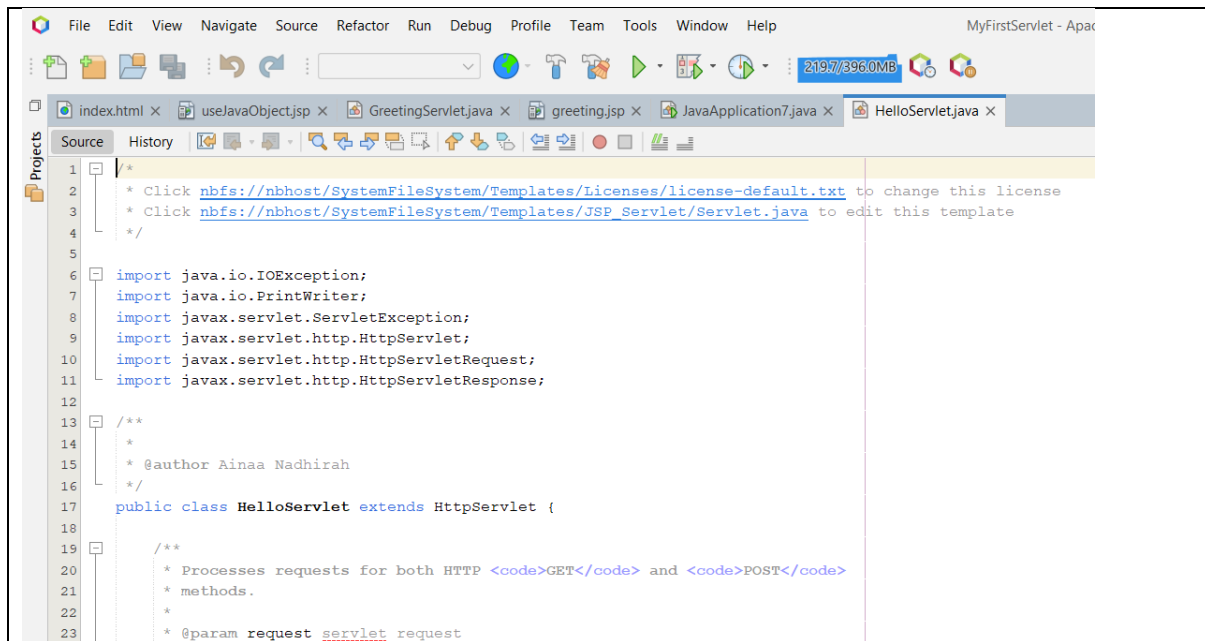
PN. WAN FATIN FATIAH BINTI YAHYA

LAB 1

Task 6: Writing a Simple Java Servlet

HelloServlet.java File

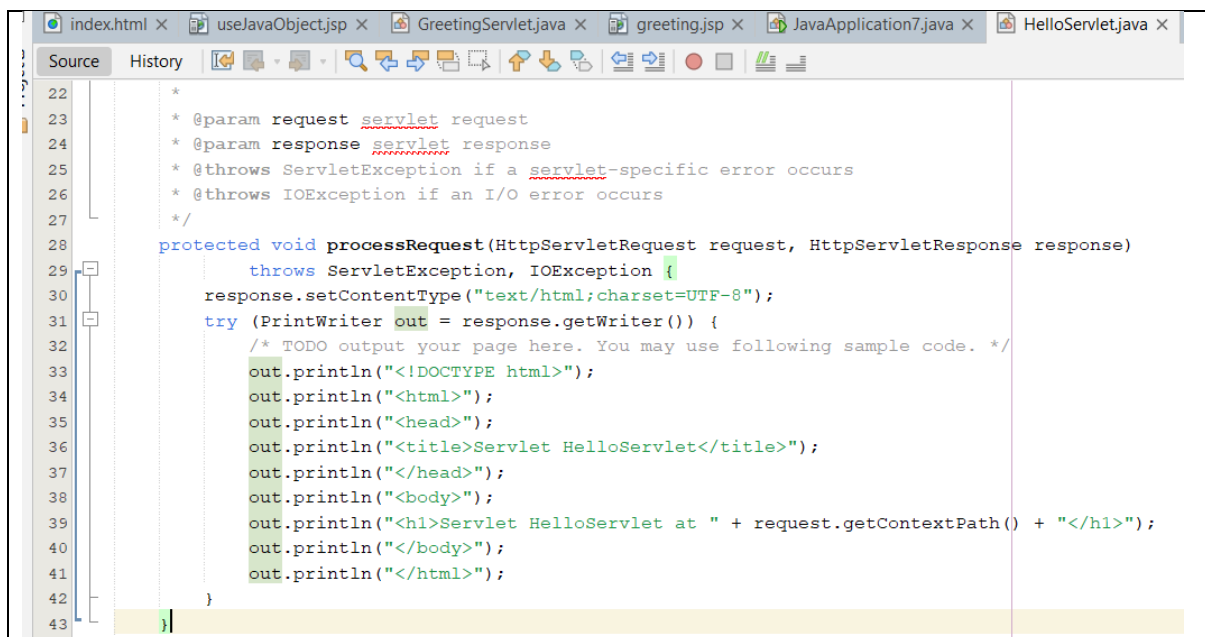
Code:



```
1  /*
2   * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license
3   * Click nbfs://nbhost/SystemFileSystem/Templates/JSP\_Servlet/Servlet.java to edit this template
4   */
5
6  import java.io.IOException;
7  import java.io.PrintWriter;
8  import javax.servlet.ServletException;
9  import javax.servlet.http.HttpServlet;
10 import javax.servlet.http.HttpServletRequest;
11 import javax.servlet.http.HttpServletResponse;
12
13 /**
14  *
15  * @author Ainaa Nadhirah
16  */
17 public class HelloServlet extends HttpServlet {
18
19     /**
20      * Processes requests for both HTTP GET and POST
21      * methods.
22      *
23      * @param request servlet request
```

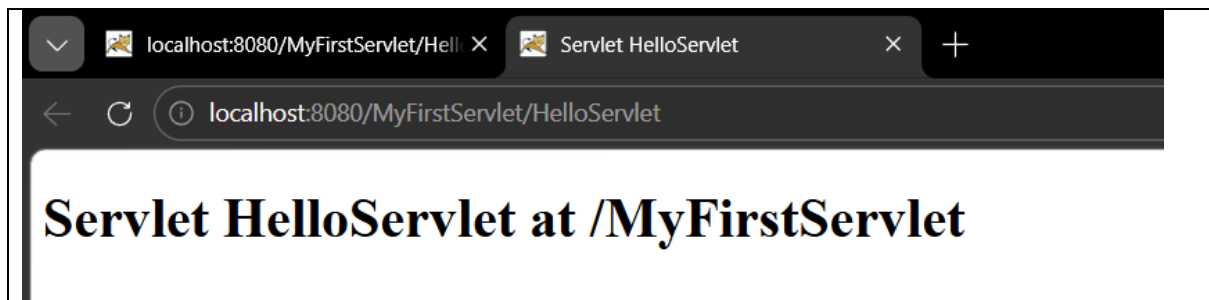
processRequest() method:

Code:



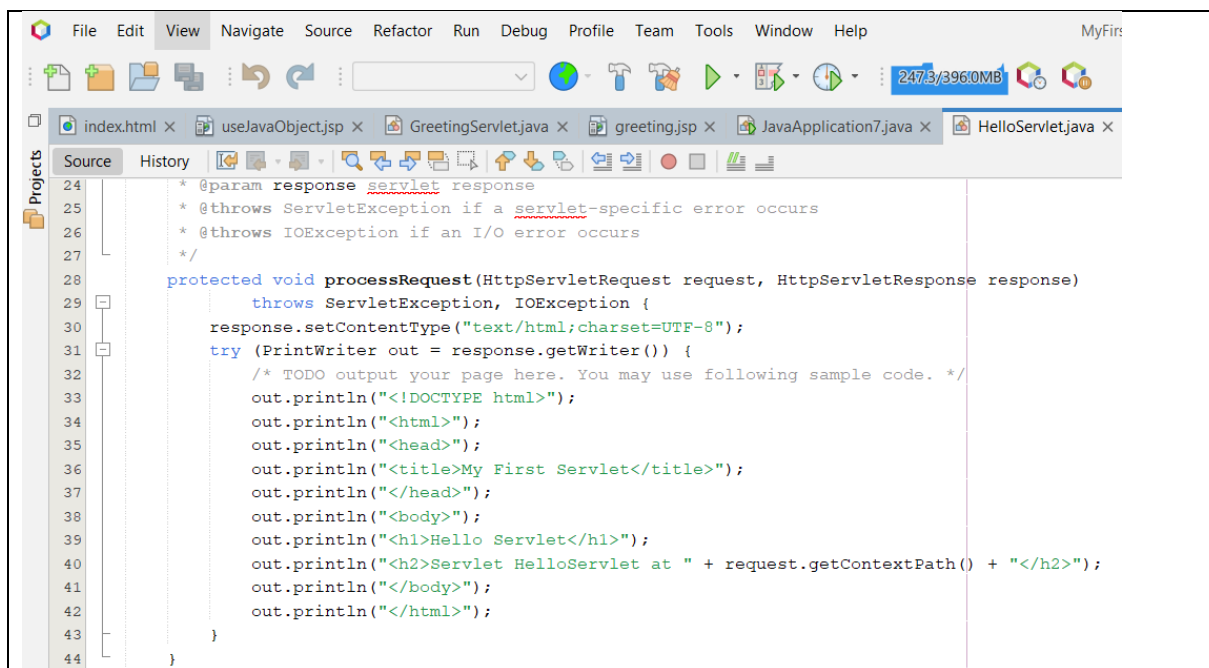
```
22     *
23     * @param request servlet request
24     * @param response servlet response
25     * @throws ServletException if a servlet-specific error occurs
26     * @throws IOException if an I/O error occurs
27     */
28     protected void processRequest(HttpServletRequest request, HttpServletResponse response)
29         throws ServletException, IOException {
30         response.setContentType("text/html;charset=UTF-8");
31         try (PrintWriter out = response.getWriter()) {
32             /* TODO output your page here. You may use following sample code. */
33             out.println("<!DOCTYPE html>");
34             out.println("<html>");
35             out.println("<head>");
36             out.println("<title>Servlet HelloServlet</title>");
37             out.println("</head>");
38             out.println("<body>");
39             out.println("<h1>Servlet HelloServlet at " + request.getContextPath() + "</h1>");
40             out.println("</body>");
41             out.println("</html>");
42         }
43     }
```

Result:

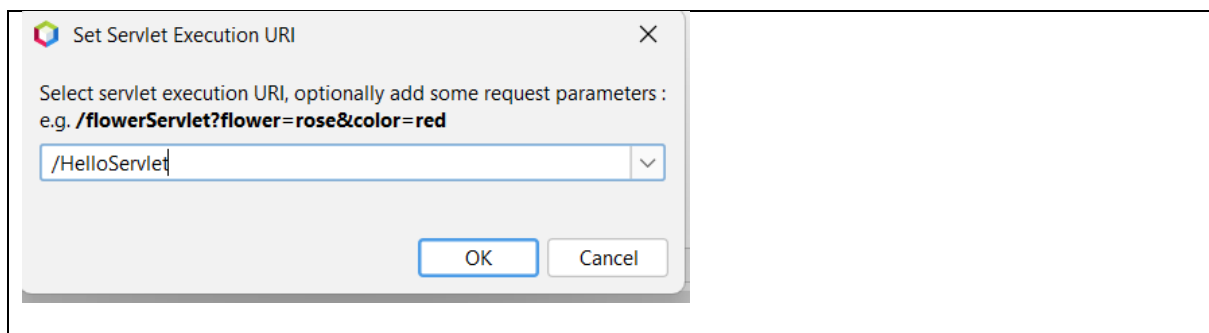


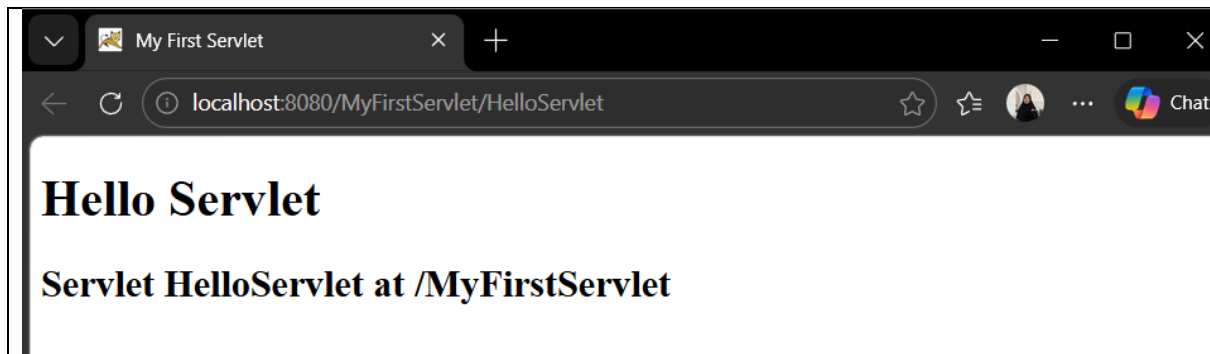
After modify processRequest() Method:

Code:



Result:



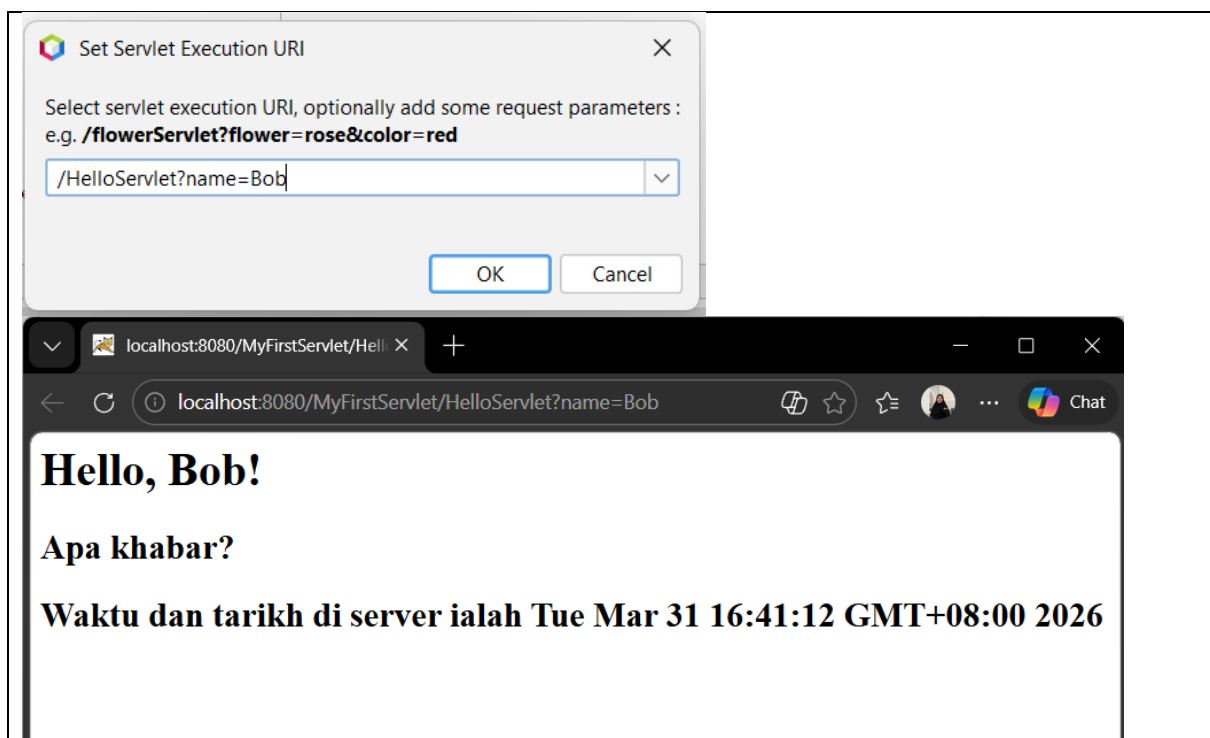


After modify the doGet() method:

Code:

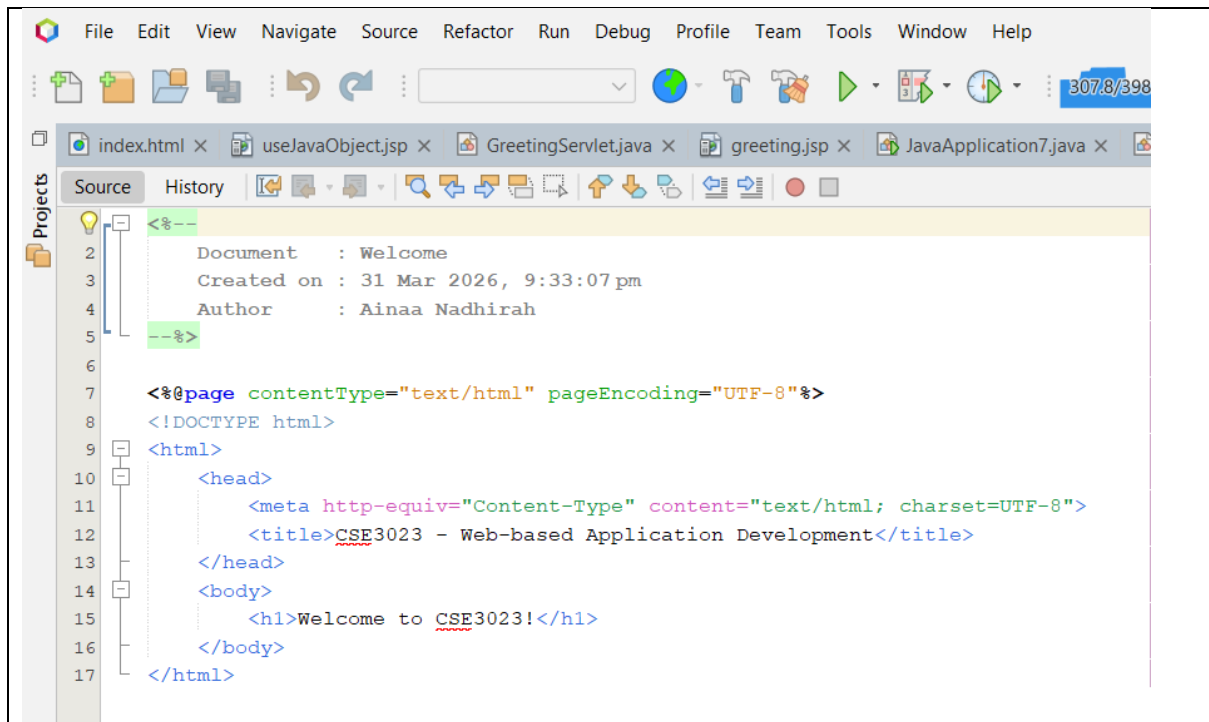


Result:



Task 7: Writing a simple JSP Program

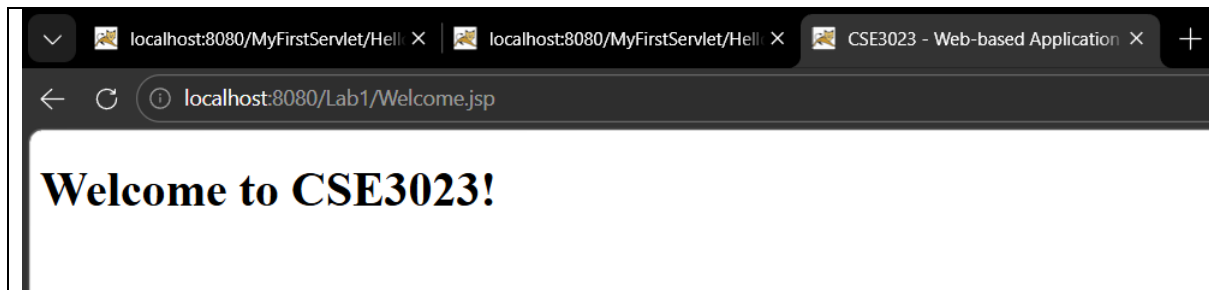
JSP File:



The screenshot shows an IDE with the following code in 'greeting.jsp':

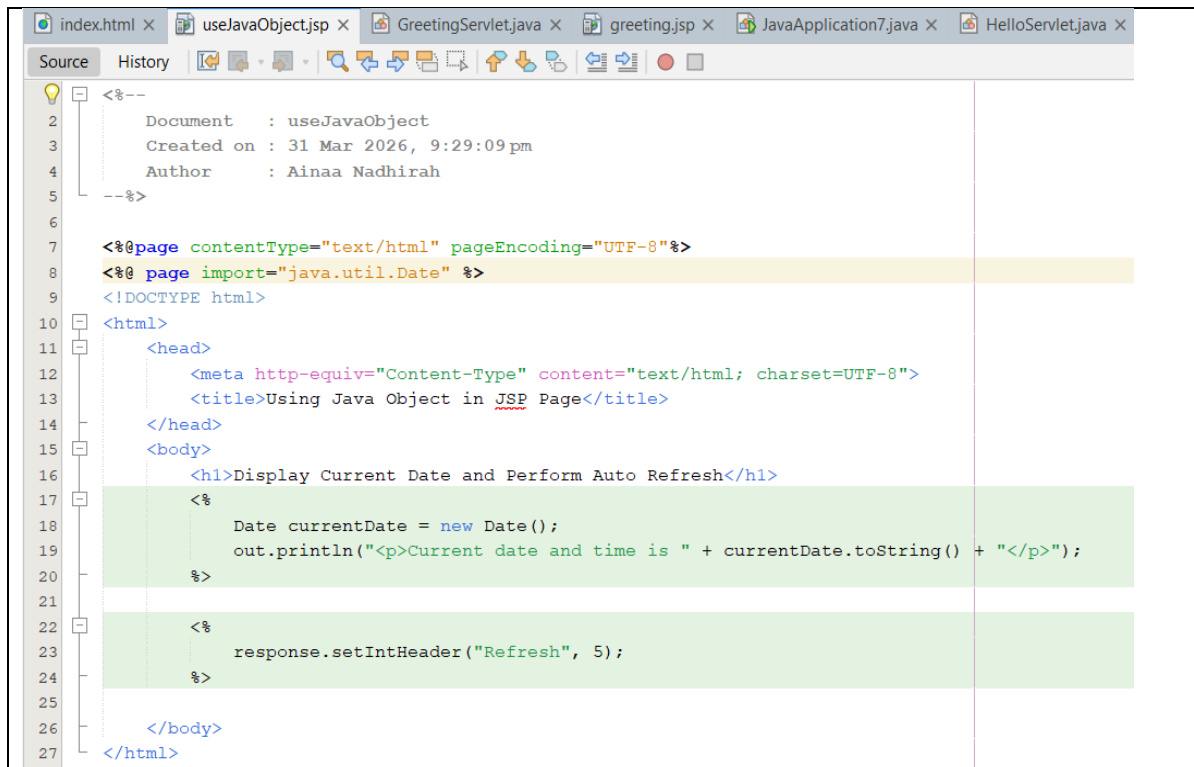
```
1 <!--
2     Document    : Welcome
3     Created on  : 31 Mar 2026, 9:33:07 pm
4     Author     : Ainaa Nadhirah
5 -->
6
7 <%@page contentType="text/html" pageEncoding="UTF-8"%>
8 <!DOCTYPE html>
9 <html>
10 <head>
11     <meta http-equiv="Content-Type" content="text/html; charset=UTF-8">
12     <title>CSE3023 - Web-based Application Development</title>
13 </head>
14 <body>
15     <h1>Welcome to CSE3023!</h1>
16 </body>
17 </html>
```

Result after running at web browser:



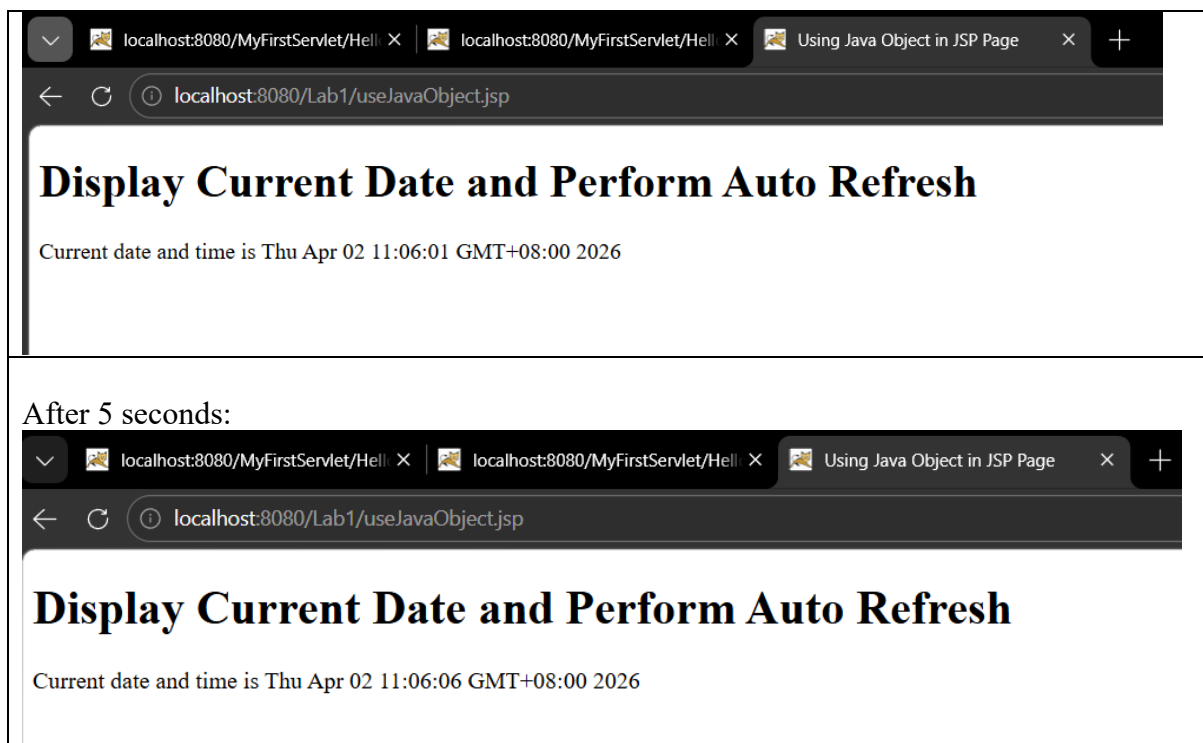
Task 8: Use Java Reference Datatype/Class Wrapper in JSP

JSP File – useJavaObject.jsp



```
1 <!--
2     Document    : useJavaObject
3     Created on  : 31 Mar 2026, 9:29:09 pm
4     Author     : Ainaa Nadhirah
5 -->
6
7 <%@page contentType="text/html" pageEncoding="UTF-8"%>
8 <%@ page import="java.util.Date" %>
9 <!DOCTYPE html>
10 <html>
11 <head>
12 <meta http-equiv="Content-Type" content="text/html; charset=UTF-8">
13 <title>Using Java Object in JSP Page</title>
14 </head>
15 <body>
16 <h1>Display Current Date and Perform Auto Refresh</h1>
17
18 <%
19     Date currentDate = new Date();
20     out.println("<p>Current date and time is " + currentDate.toString() + "</p>");
21 %>
22
23 <%
24     response.setIntHeader("Refresh", 5);
25 %>
26
27 </body>
28 </html>
```

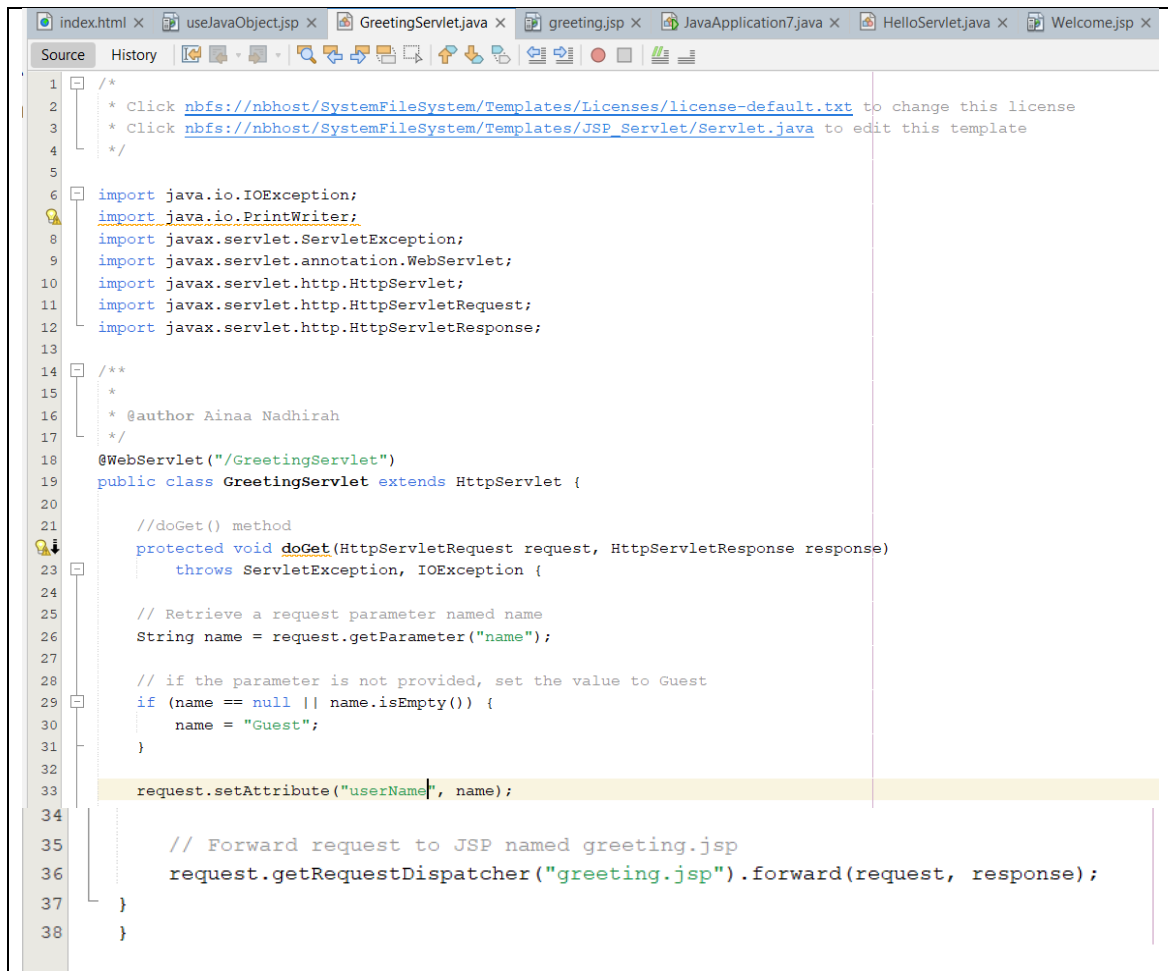
Result:



Exercise:

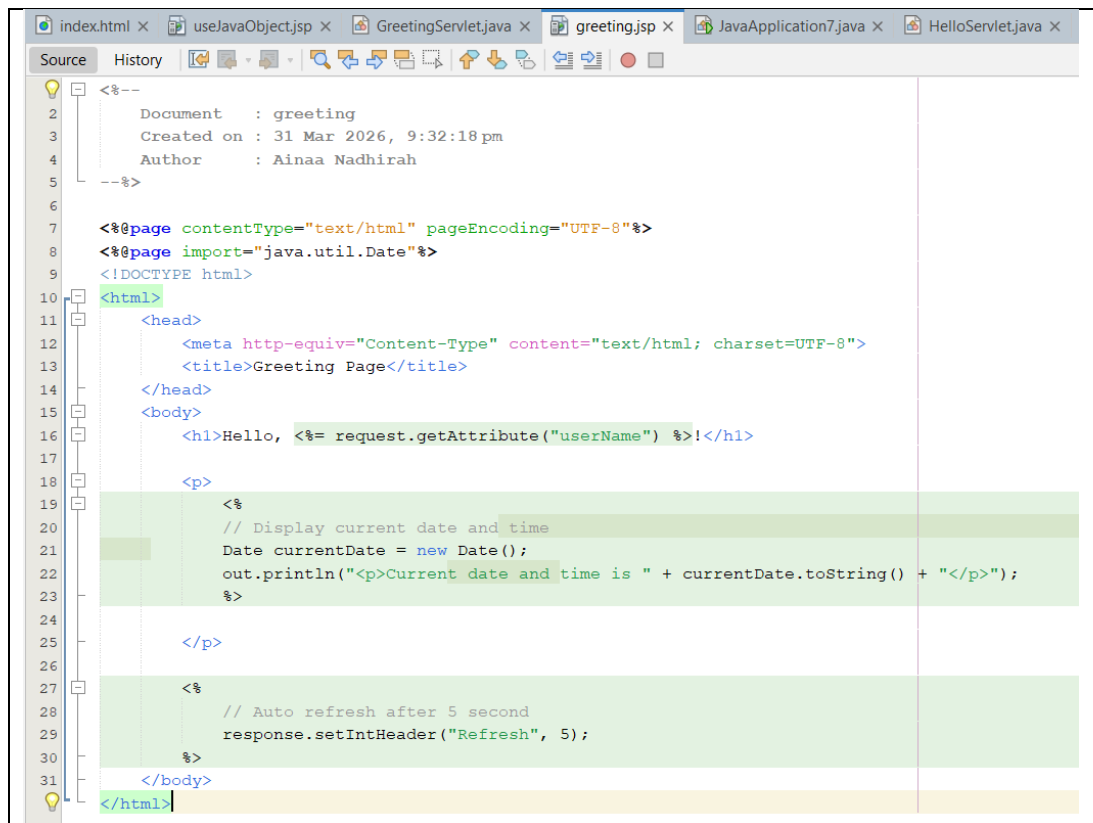
Code:

GreetingServlet.java



```
1  /*
2  * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license
3  * Click nbfs://nbhost/SystemFileSystem/Templates/JSP\_Servlet/Servlet.java to edit this template
4  */
5
6  import java.io.IOException;
7  import java.io.PrintWriter;
8  import javax.servlet.ServletException;
9  import javax.servlet.annotation.WebServlet;
10 import javax.servlet.http.HttpServlet;
11 import javax.servlet.http.HttpServletRequest;
12 import javax.servlet.http.HttpServletResponse;
13
14 /**
15 *
16 * @author Ainaa Nadhirah
17 */
18 @WebServlet("/GreetingServlet")
19 public class GreetingServlet extends HttpServlet {
20
21     //doGet() method
22     protected void doGet(HttpServletRequest request, HttpServletResponse response)
23         throws ServletException, IOException {
24
25         // Retrieve a request parameter named name
26         String name = request.getParameter("name");
27
28         // if the parameter is not provided, set the value to Guest
29         if (name == null || name.isEmpty()) {
30             name = "Guest";
31         }
32
33         request.setAttribute("userName", name);
34
35         // Forward request to JSP named greeting.jsp
36         request.getRequestDispatcher("greeting.jsp").forward(request, response);
37     }
38 }
```

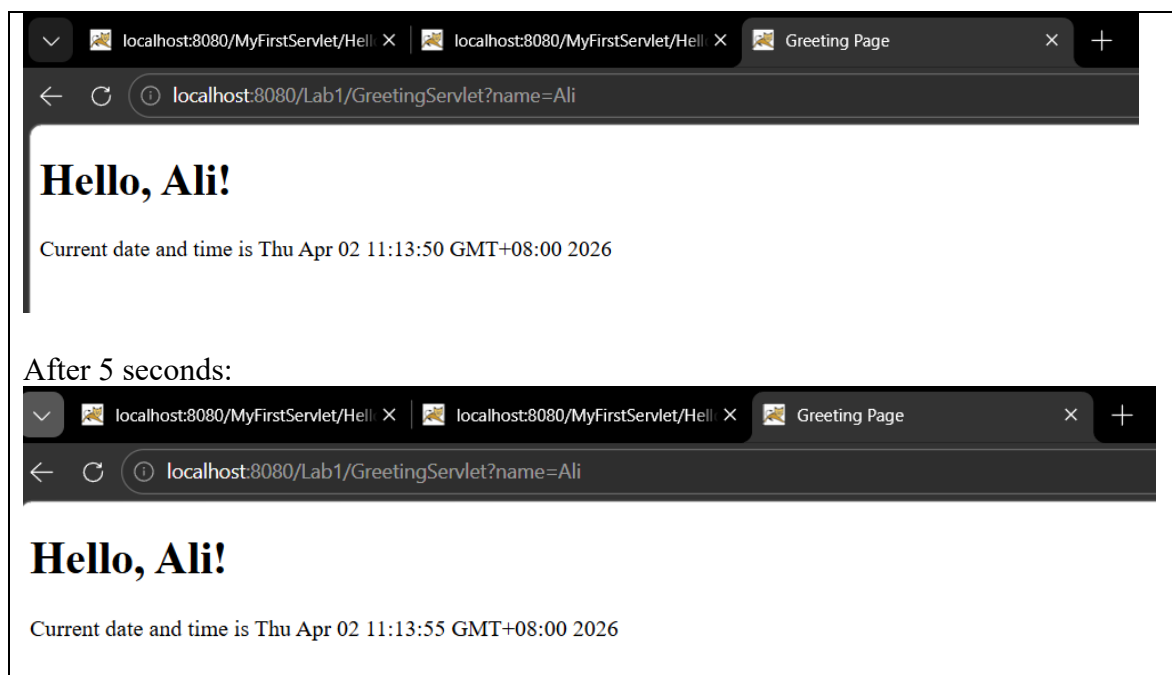
greeting.jsp



```
1  <!--
2      Document   : greeting
3      Created on : 31 Mar 2026, 9:32:18 pm
4      Author    : Ainaa Nadhirah
5  -->
6
7  <%@page contentType="text/html" pageEncoding="UTF-8"%>
8  <%@page import="java.util.Date"%>
9  <!DOCTYPE html>
10 <html>
11 <head>
12 <meta http-equiv="Content-Type" content="text/html; charset=UTF-8">
13 <title>Greeting Page</title>
14 </head>
15 <body>
16 <h1>Hello, <%= request.getAttribute("userName") %>!</h1>
17
18 <p>
19 <%
20 // Display current date and time
21 Date currentDate = new Date();
22 out.println("<p>Current date and time is " + currentDate.toString() + "</p>");
23 %>
24
25 </p>
26
27 <%
28 // Auto refresh after 5 second
29 response.setIntHeader("Refresh", 5);
30 %>
31 </body>
32 </html>
```

Result:

When name= Ali



When the user didn't enter the name:

